

ELECTRICAL & ELECTRONICS ENGINEERING

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
EET303	MICROPROCESSORS AND MICROCONTROLLERS	PCC	3	1	0	4

Preamble: This course helps the students to understand 8085 microprocessor and 8051 microcontroller architecture as well as to design hardware interfacing circuit. This also aids to thrive their programming skills to solve real world problems.

Prerequisite: Fundamentals of Digital Electronics, C Programming

Course Outcomes: After the completion of the course the student will be able to:

CO 1	Describe the architecture and timing diagram of 8085 microprocessor.
CO 2	Develop assembly language programs in 8085 microprocessor.
CO 3	Identify the different ways of interfacing memory and I/O with 8085 microprocessor.
CO 4	Understand the architecture of 8051 microcontroller and embedded systems.
CO 5	Develop assembly level and embedded C programs in 8051 microcontroller.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	2										
CO 2	3	2	3	2	1							
CO 3	3	2	2	2	2							
CO 4	3	2										
CO 5	3	2	3	2	1	1						1

Assessment Pattern:

Total Marks	CIE marks	ESE marks	ESE Duration
150	50	100	03 Hrs

Bloom's Category	Continuous Assessment Tests		End Semester Examination
	1	2	
Remember (K1)	10	10	20
Understand (K2)	10	10	20
Apply (K3)	30	30	60

ELECTRICAL & ELECTRONICS ENGINEERING

Analyse (K4)			
Evaluate (K5)			
Create (K6)			

End Semester Examination Pattern : There will be two parts; Part A and Part B.

Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions.

Part B contains 2 questions from each module of which student should answer any one. Each question carries 14 marks and can have maximum 2 sub-divisions.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Describe the register organization in 8085 microprocessor.
2. Explain the Stack and subroutine operations.
3. Explain the basic steps involved in accessing memory locations.
4. Draw the timing diagrams of different instructions of 8085 microprocessor.

Course Outcome 2 (CO2):

1. Describe the addressing modes of 8085 microprocessor.
2. Describe the various types of 8085 microprocessor instructions.
3. Explain in detail the instruction set of 8085 microprocessor.
4. Write an ALP for data transfer, arithmetic, logical and branching operations.

Course Outcome 3(CO3):

1. Explain how RAM and ROM memory are interfaced with 8085 microprocessor.
2. Describe address decoding used in I/O interfacing.
3. Explain the architecture of 8255 PPI.
4. Explain the modes of operation of 8255 PPI.

Course Outcome 4 (CO4):

1. Explain the special function registers in 8051 microcontroller.
2. Explain the operating modes of serial port of 8051 microcontroller.
3. Describe the addressing modes and modes of operation of timer of 8051 microcontroller.
4. Explain the embedded C Programming.

Course Outcome 5 (CO5):

1. Explain timer programming in assembly language and embedded C.
2. Explain serial port programming in assembly language and embedded C.
3. How to interface ADC, DAC and sensors with 8051 microcontroller.
4. Explain interrupt programming in assembly language and C.

Model Question Paper

QP Code:

Pages: 2

Reg No: _____

Name: _____

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIFTH SEMESTER B.TECH DEGREE EXAMINATION,
MONTH & YEAR**

Course Code: EET303

Course Name: MICROPROCESSORS AND MICROCONTROLLERS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all Questions. Each question carries 3 Marks

1. Explain the use of ALE signal in Intel 8085 microprocessor.
2. Describe the use of CLK OUT and RESET OUT signals.
3. With the help of an example explain the operation of XTHL instruction.
4. How can we check the status of flags in 8085 microprocessor?
5. Explain software and hardware interrupts.
6. Write the differences between microprocessor and microcontroller.
7. Draw the block diagram of 8051 microcontroller.
8. Explain the bit pattern of TMOD register of 8051 microcontroller.
9. How we can enable and disable interrupts in 8051 microcontroller.
10. Find the bits of TMOD registers to operate as timers in the following modes
(i) Mode 1 Timer (ii) Mode 2 Timer 0.

PART B

Answer any one full question from each module. Each question carries 14 Marks

Module 1

11. (a) Explain the functional block diagram of 8085 microprocessor. (10)
(b) Define machine cycle and T state. (4)
12. (a) Sketch and explain the timing diagram of LDA 2003H. (10)
(b) Describe the addressing modes of 8085 microprocessor. (4)

Module 2

13. (a) Write an ALP to sort an array of 10 numbers stored from memory location 2001H onwards in ascending order. (10)

(b) Explain stack related operations in 8085 microprocessor. (4)

14. (a) Write a delay program to introduce a delay of 1 second. (8)

(b) Explain the operation of DAA instruction in 8085 microprocessor. (6)

Module 3

15. (a) Explain the address decoding technique in memory interfacing. (8)

(b) Give the control word format for BSR and I/O Mode in 8255. (6)

16. (a) Explain the architecture of 8051 microcontroller. (8)

(b) Explain hard and soft real time systems. (6)

Module 4

17. (a) Explain the different methods to create a time delay in 8051 microcontroller. (7)

(b) Explain the different addressing modes of 8051 microcontroller? (7)

18. (a) Explain the various types of instructions in 8051 microcontroller? (6)

(b) Write a Program in 8051 for the generation of square wave having a duty ratio of 0.5 for a time period of 1ms. (8)

Module 5

19. (a) Explain how a DAC can be interfaced to 8051 microcontroller. (10)

(b) Explain the role of SBUF and SCON registers used in 8051 microcontroller. (4)

20. (a) Describe the generation of time delay using the timer of 8051 microcontroller. (8)

(b) Explain the various interrupts in 8051 microcontroller. (6)

Syllabus

Module 1

Internal architecture of 8085 microprocessor–Functional block diagram

Instruction set-Addressing modes - Classification of instructions - Status flags.

Machine cycles and T states – Fetch and execute cycles- Timing diagram for instruction and data flow.

Module 2

Introduction to assembly language programming- Data transfer operations, arithmetic operations, logic operations, branching operations, I/O and machine control operations.

Assembly language programmes (ALP) in 8085 microprocessor- Data handling/Data transfer, Arithmetic operations, Code conversion- BCD to Binary - Binary to BCD, Sorting - Ascending and descending including bubble sorting.

Stack and subroutines – Conditional CALL and Return instructions

Time delay subroutines using 8 bit register, 16 bit register pair and Nested loop control.

Module 3

Interrupt & interrupt handling - Hardware and Software interrupts.

I/O and memory interfacing – Address decoding– Interfacing I/O ports -Programmable Peripheral Interface PPI 8255 - Modes of operation- Interfacing of seven segment LED.

Introduction to embedded systems, Current trends and challenges, Applications of embedded systems- Hard and soft real time systems.

Introduction to microcontrollers- Microprocessor Vs Microcontroller- 8051 Microcontrollers – Hardware - Microcontroller architecture and programming model - I/O port structure - Register organization -General purpose RAM - Bit addressable RAM - Special Function Registers (SFRs).

Module 4

Instruction set - Instruction types - Addressing modes of 8051 microcontrollers.

8051 microcontroller data types and directives - Time delay programmes and I/O port programming.

Introduction to embedded C Programming - time delay in C - I/O port programming in embedded C.

Module 5

8051 Timer/counter programming - Serial port programming - Interrupt programming in assembly language and embedded C.

Interfacing –ADC - DAC and temperature sensor

Text Books

1. Ramesh Gaonkar, “Microprocessor Architecture Programming and Applications”, Penram International Publishing; Sixth edition, 2014.
2. Mohamed Ali Mazidi, Janice GillispieMazidi, “The 8051 microcontroller and embedded systems using Assembly and C”, second edition, Pearson/Prentice hall of India.
3. Kenneth J. Ayala, “The 8051 microcontroller”, 3rd edition, Cengage Learning, 2010
4. Lyla B Das, “Embedded Systems - An Integrated Approach”, Pearson Education India

Reference Books

1. B Ram, “Fundamentals of Microprocessors and Microcontrollers”, 9e, DhanpatRai Publications, 2019.
2. Wadhwa, “Microprocessor 8085 microprocessor: Architecture, Programming and Interfacing”, PHI 2010
3. Shibu K V, “Introduction to Embedded systems”, TMH

Course Contents and Lecture Schedule

No.	Topic	No. of Lectures
1	Architecture and Instruction set of 8085 microprocessor (9 hours)	
1.1	Internal architecture of 8085 microprocessor– functional block diagram	2
1.2	Instruction set- Addressing modes, Classification of instructions - Status flags.	4
1.3	Machine cycles and T states – Fetch and execute cycles - timing diagram for instruction and data flow.	3
2	Assembly language programming (9 hours)	
2.1	Introduction to assembly language programming- data transfer operations, arithmetic operations, logic operations, branching operations, I/O and machine control operations.	2
2.2	Assembly language programmes (ALP) in 8085 microprocessor-Data handling/Data transfer - Arithmetic operations - Code conversion - BCD to Binary - Binary to BCD, Sorting - Ascending and descending including bubble sorting.	4

ELECTRICAL & ELECTRONICS ENGINEERING

2.3	Stack and subroutines – Conditional call and return instructions – Stack operations.	2
2.4	Time delay subroutines using 8bit register, 16 bit register pair and Nested loop control.	1
3	Interfacing circuits for 8085 microprocessor and introduction to 8051 Microcontroller (10 hours)	
3.1	Interrupt and interrupt handling - Hardware and Software interrupts.	1
3.2	I/O and memory interfacing – Address decoding – Interfacing I/O ports-Programmable peripheral interface PPI 8255 - Modes of operation -Interfacing of seven segment LED.	4
3.3	Introduction to embedded systems - Current trends and challenges - Applications of embedded systems - Hard and Soft real time systems.	1
3.4	Introduction to microcontrollers - Microprocessor Vs Microcontroller - 8051- Microcontrollers - Hardware	1
3.5	Microcontroller Architecture and programming model: I/O Port structure - Register organization - General purpose RAM -Bit Addressable RAM -Special Function Registers (SFRs).	3
4	Programming of 8051 Microcontroller (9 hours)	
4.1	Instruction Set - Instruction Types - Addressing modes	3
4.2	8051- Data types and directives -Time delay programmes and I/O port programming.	3
4.3	Introduction to embedded C Programming - Time delay in C - I/O port programming in embedded C.	3
5	Interfacing circuits of 8051 Microcontroller (9 hours)	
5.1	Timer/counter programming in assembly language and embedded C	3
5.2	Serial port programming in assembly language and embedded C	2
5.3	Interrupt programming in assembly language and embedded C	2
5.4	Interfacing –ADC - DAC and temperature sensor	2